

Lead-to-Revenue Automation Pipeline

8-Signal Scoring, Multi-CRM Sync, ML Revenue Prediction

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Project Overview

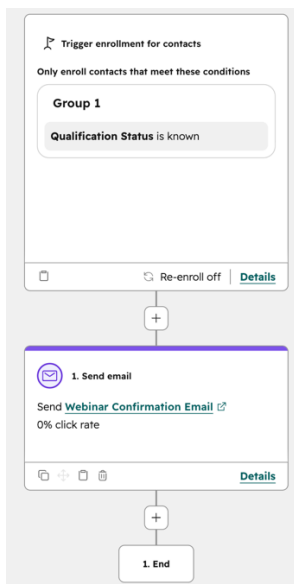
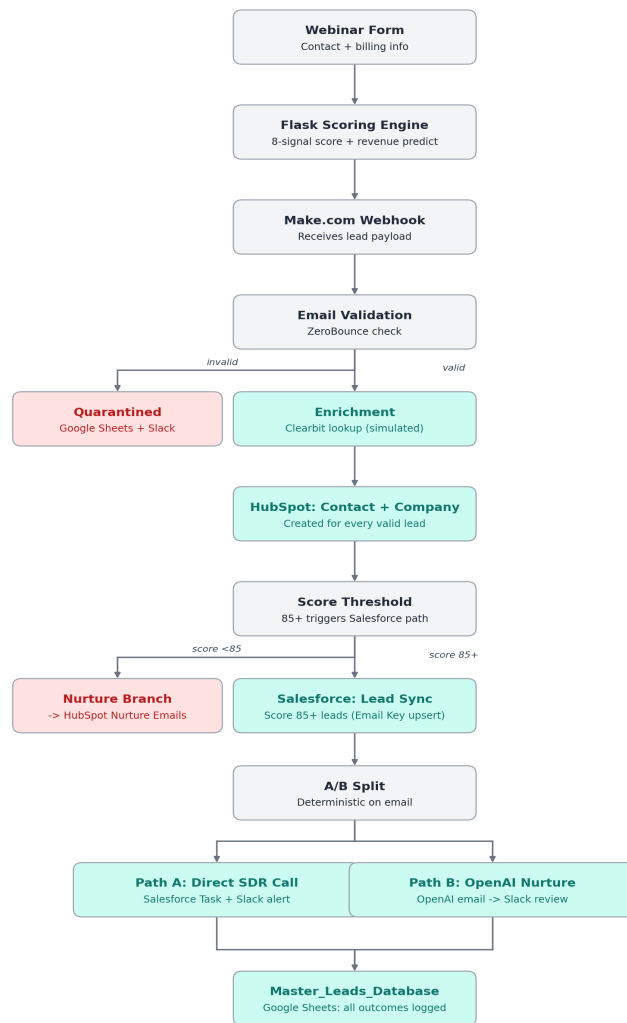
Built an automated lead scoring + revenue prediction system for **Bill Swift**, a fictional usage-based billing company modelled on Chargebee

- Every webinar signup gets scored out of 100 based on real buying signals (job title, company size, billing volume, etc.)
- Scoring logic built in **Python (Flask)** — a trained **Random Forest ML model** predicts how much revenue each lead is worth
- **ZeroBounce** verifies email deliverability before anything touches the CRM
- **Make.com** orchestrates the entire pipeline across two automation scenarios
- Every valid lead syncs to **HubSpot**; high-scoring leads also sync to **Salesforce**
- High-scoring leads get an immediate sales call (Salesforce Task + **Slack** alert) or a personalized email drafted by **OpenAI**
- Lower-scoring leads get automated nurture emails via native **HubSpot Workflows** — booking a call upgrades them into the same high-touch path
- Every outcome — every branch, every path — is logged to a shared **Google Sheet** for reporting

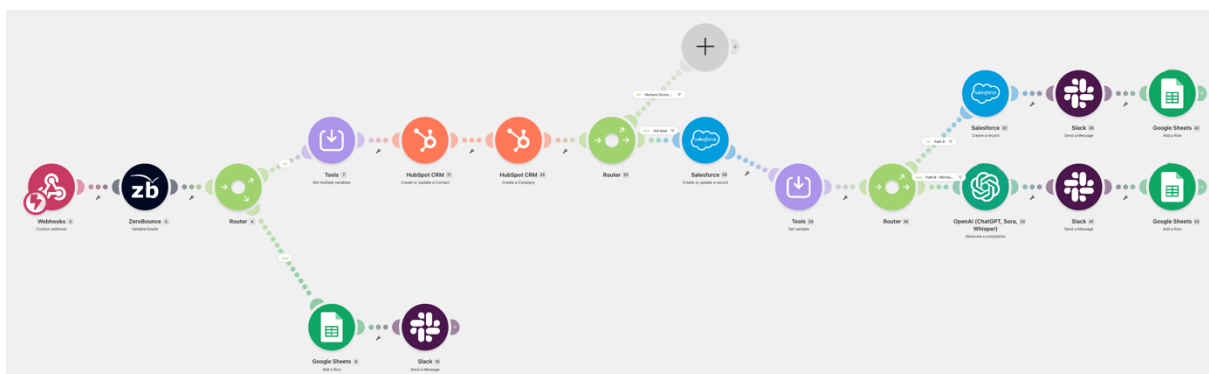
Stack: Python, Make.com, HubSpot, Salesforce, ZeroBounce, OpenAI, Google Sheets, Slack

Pipeline – Scenario 1

Form submission → scoring → validation → CRM sync → score gate → A/B-tested outreach.



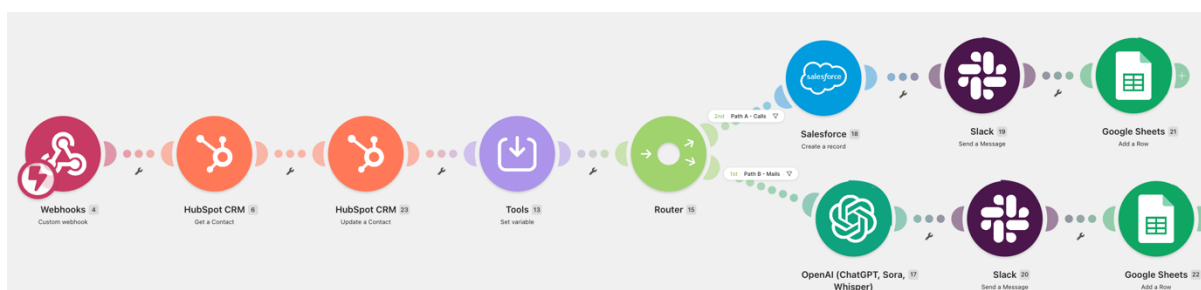
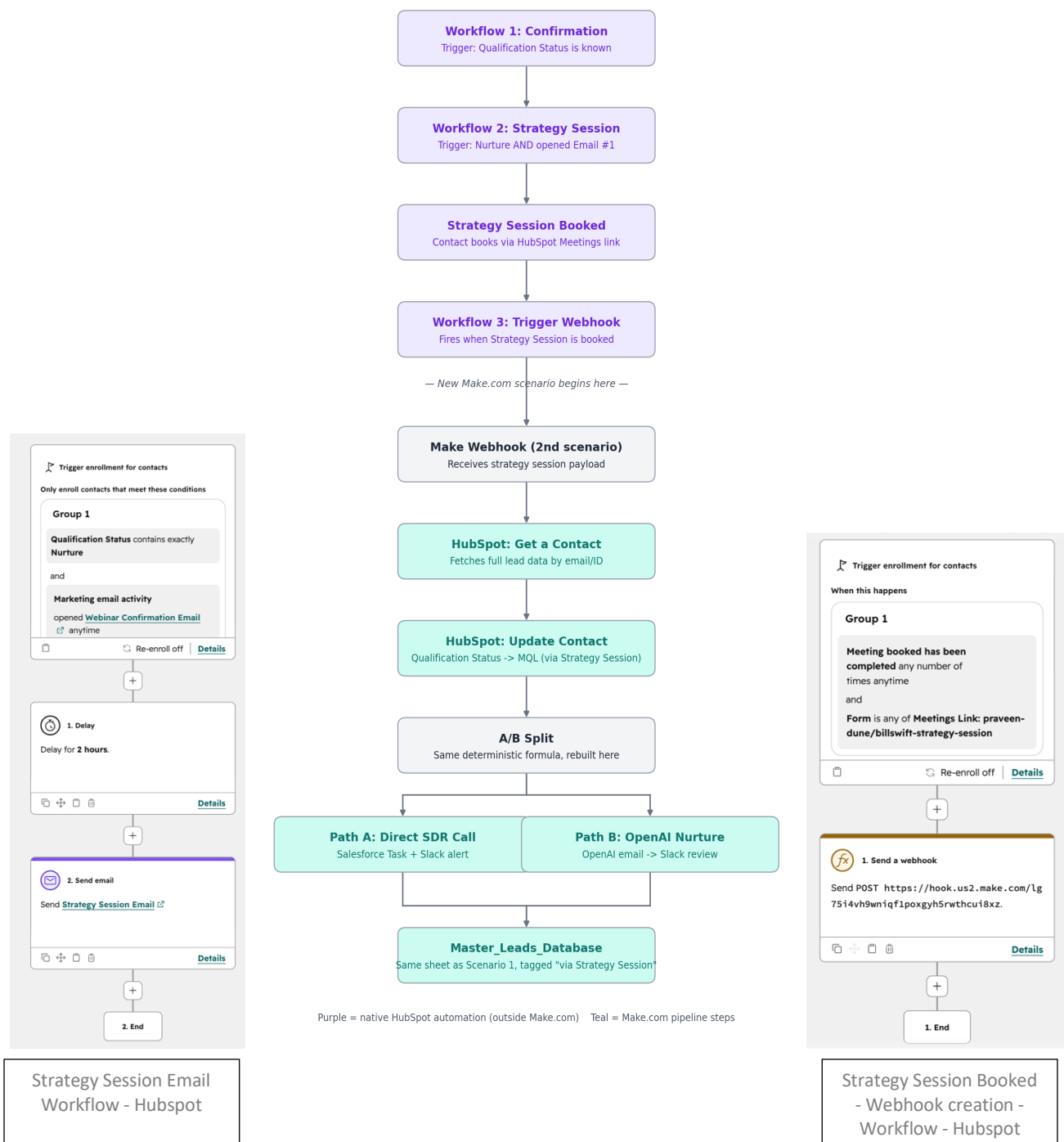
Webinar Confirmation
Mail Workflow



Make.com Scenario 1

Pipeline – Scenario 2

Low-score leads aren't dropped. If they engage enough to book a strategy session, that real behavior earns them the same high-touch treatment as a top-scoring lead.



Make.com Scenario 2

1. Scoring Engine — 8 Signals, 100 Points

Computed server-side (tamper-proof). Every weight is grounded in a real Chargebee fact, not a guess.

Signal	Points	Real-World Grounding
Monthly Billing Volume	20	Companies processing \$100K+/month have hit Chargebee's real Enterprise-plan threshold
Primary Revenue Challenge	20	Matches the exact problems covered in the real webinar content
Current Billing Tool	12	Matches the exact problems covered in the real webinar content
Job Title	12	Chargebee is finance-led — CFO/RevOps are the real buyers
Company Size	12	Mirrors Chargebee's real customer growth stages, from startup to enterprise
Email Domain	8	Personal email = colder lead
CRM Tech Stack	8	Chargebee natively integrates with Salesforce and HubSpot
Industry	8	Over half of Chargebee's real customers are software and IT companies

Cold 0–35 · Warm 36–65 · Hot 66–100 · MQL gate at 85+

- If someone selects **"Other"** on a dropdown, that field scores 0 points, no matter what they type — this applies to 4 fields on the form
- A machine learning model predicts how much revenue each lead is worth, trained on 250 rows of simulated past deals modeled on Chargebee's real pricing plans, deliberately balanced so about 25% of leads come out fully qualified
- That predicted revenue number gets saved to the Google Sheet alongside every lead's score — it's not just used internally by the model and thrown away.

2. Validation & Quarantine

- ZeroBounce checks deliverability before any CRM write
- Invalid leads → Google Sheets quarantine log + real-time Slack alert
- Personal emails still pass — flagged in scoring, not blocked

3. Multi-CRM Sync

- Every valid lead gets added to HubSpot, no matter their score
- Leads scoring 85+ also get added to Salesforce — if they've submitted before, their existing record updates instead of creating a duplicate
- Company records in HubSpot use just the company name to identify them, not the website domain — using domain caused errors for leads with personal emails (like Gmail), since there's no real company domain to match
- Every Salesforce Task (the "call this lead" reminder) is always created fresh — never reused or updated from an old one

4. Score Gate & A/B Test

- 85+ leads split deterministically by email (not random) into two paths
- Path A — Direct SDR Call: Salesforce Task ("Call in 15 min") + Slack alert
- Path B — OpenAI Nurture: AI drafts a personalized email → posted to Slack for human review before sending.

5. Native HubSpot Nurture Sequence

For leads that don't score high enough right away, three automated HubSpot emails keep them engaged:

- When a lead is added, they automatically get a webinar confirmation email
- If they open that email, they get a second email offering a private strategy call, with a link to book it
- If they actually book that call, it triggers the next stage of automation — Make's connector doesn't support this trigger natively, so a webhook bridges HubSpot to the automation
- That lead's status upgrades to qualified, and they now get the same treatment as a lead who scored high initially — booking a real call matters just as much as a good form score

6. Metrics

Every outcome — no matter which path a lead takes — gets logged in one shared Google Sheet. That single sheet is what lets you calculate:

- **Data Quality Rate:** what percentage of submissions were real, valid leads, not bounced or fake emails ($\text{Valid Leads} \div \text{Total Submissions}$)
- **Total Pipeline Value:** since predicted revenue is logged for every lead, you can add up expected revenue by path, not just count how many leads there were
- **A/B Conversion Rate:** which outreach method — direct calls or AI emails — actually closes more deals ($\text{Closed-Won Rate, Path A vs. Path B}$)
- **Nurture-to-Booking Rate:** what percentage of nurtured leads actually go on to book a strategy call ($\text{Bookings} \div \text{Workflow 2 Enrolments}$)

Conclusion

This project began as a simple lead scoring script and developed into a complete system: two automated workflows, three integrated CRM and marketing platforms, a trained machine learning model, and a genuine A/B test. Each scoring weight is grounded in real Chargebee data, and each technical decision was made in response to a real problem encountered during development — not assumed in advance. Together, they demonstrate the ability to design a system end to end: understanding the business need, translating it into sound logic, and implementing it across real platforms.